Facial visibility and the emergence of communication systems: Insights from behavioral measures and facial signals

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The primary mode of (early) human communication is face-to-face interaction, involving multimodal and meta-communicative signals, including facial signals (Holler, 2025; Vigliocco et al., 2014). It is now widely recognized that facial signals can support communication in existing languages by e.g. facilitating the prediction of upcoming utterances, repair, backchanneling, and signalling uncertainty (Emmendorfer & Holler, 2025; Hömke et al., 2025; Micklos & Woensdregt, 2023; Nölle et al., 2023; Bavelas & Chovil, 2018; Hömke et al., 2018; Swerts & Krahmer, 2005). These types of processes likely also contribute to the pragmatic inferences relevant to coordinating on a new communication system (Healey et al., 2007; Micklos et al., 2018; Roberts et al., 2018). However, communication experiments that simulate the process of coordinating on new communication systems often overlook the role of such signals. Even more strikingly, many are not conducted face-to-face. While experimental work is scarce, theories suggest the face played a crucial role in language evolution, mainly in facilitating the transition between a gesture-dominant and a vocalization-dominant system (see Wacewicz et al. (2016) for an overview). The current work aims to contribute to this area of research by investigating the role of the facial signals in the emergence of conventionalized symbol systems more broadly, and how they can serve as meta-communicative cues to help establish common ground in interaction.

To achieve this, we make use of a communication game paradigm where we manipulate facial visibility in a between-subjects design (34 dyads total) with two conditions: face visible and face invisible. Dyads repeatedly communicate about twelve shapes (adapted from Macuch Silva et al. (2020)) while being video and audio recorded, with roles switching every trial: the producer writes a novel label for the target shape, and the guesser guesses the target shape upon seeing the label. After guessing, both participants get feedback about the target and guessed shape. Any communication using existing language was prohibited. Participants played four rounds using the same shapes.

Participants who did not see each other lacked access to facial signals that support pragmatic inference, for example by signalling understanding or trouble in understanding (Hömke et al., 2018, 2025), but also confidence in a linguistic label or guess (Nölle et al., 2023; Swerts & Krahmer, 2005). We predicted that this lack of information would decrease communicative success and convergence. However, we also acknowledge that participants might compensate with strategies that support communication. We analyzed three aspects: (1) communicative success, measured as accuracy of a guess, and (2) convergence, capturing how similarly participants in a dyad label a shape, and (3) eyebrow movements, for which segments of the interaction (producing the label, presenting the label, and guessing) have been annotated binarily for each participant in ELAN (Sloetjes & Wittenburg, 2008). These annotations were then summed to investigate associations between the amount of eyebrow movement in a trial and communicative success, convergence, and linguistic adaptation.

Results of mixed effects models showed communicative success and convergence increasing over rounds ($p < .001$), but no condition effect. However, the face invisible condition did show more variance in convergence ($p < .001$): pairs who could see each other converged more uniformly, while the pairs in the face invisible condition varied more. Trials had less eyebrow movement when communication was successful ($p < .01$) and convergence was higher ($p < .01$). In both conditions, labels changed more after trials in which labeling ($p < .05$) or guessing ($p < .001$) took longer, or there was more eyebrow movement ($p < .05$), which tended to go hand-in-hand ($p < .001$).

The greater variation in convergence in the face invisible condition suggests that emerging communication systems are more susceptible to variation when facial signals are unavailable. Still, the similarities in communicative success and convergence between conditions suggest that facial signals are not indispensable. The eyebrow movement annotations indicate that eyebrow movements are related to uncertainty, and that these signals could have been used as cues for adaptation.

The absence of a condition effect, together with eyebrow movement and longer label/guess times going hand-in-hand, indicates that participants could rely on label/guess time for label adaptation in both conditions, while those with visible faces could use eyebrow movements as well. Greater convergence variability in the face invisible condition may thus reflect differences in how effectively participants used label/guess time alone as a cue for adapting their labels. In contrast, in the face visible condition, eyebrow movements might have served as an additional, redundant cue, providing robustness and potentially reducing convergence variability (Shannon, 1948). Taken together, these findings point towards facial signals being able to influence early communication, but not being essential for building communication systems, as participants can flexibly rely on available cues depending on context.

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